MIL-R-17852C(AS)
12 February 1982
SUPERSEDING
MIL-R-17852B(AS)
10 October 1967

MILITARY SPECIFICATION

REDUCERS, OXYGEN PRESSURE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for two types of oxygen pressure reducers.

1.2 Classification. Reducers covered by this specification shall be of the following types, as specified (see 6.2).

Type,I - On - Off
Type II - Emergency

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein,

SPECIFICATIONS

FEDERAL

O-T-236	Tetrachloroethylene (Perchloroethyle), Technical Grade
O-T-634	Trichlorethylene, Technical
BB-A-1034	Air, Compressed, for Breathing Purposes
BB-N-411	Nitrogen, Technical
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-cleated
PPP-B-636	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1660

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SPECIFICATIONS (Continued)

FEDERAL (Continued)

PPP-B-640	Box, Fiberboard, Corrugated,. Triple-wall
PPP-T-60	Tape, Packaging, Waterproof

MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-B-22191	Barrier Material, Transparent, Flexible, Heat Sealable
MIL-O-27210	Oxygen, Aviator's Breathing Liquid and Gas
MIL-T-27730)	Tape, Antiseize, Tetrafluosoethylene, with Dispenser

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-889	Dissimilar Metals

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4,3 and 6.4).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall deselected in accordance with MIL-STD-143.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Metal parts. All metal parts shall be of a corrosion resistant material or treated in a manner to render them adequately resistant to corrosion.

3.3.1.1 Dissimilar metals. Dissimilar metals, such as defined in MIL-STD-889, shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion with protective coatings.

3.3.2 Nonmetallic materials. Any nonmetallic material that is adversely affected by continued use with oxygen shall not be used.

3.3.2.1 Age. Elastomer components, except silicone, shall not be more than 12 months old from the cure date of manufacture to the date of delivery to any Government Service or to any airframe or accessory manufacturer.

3.3.2.2 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a furigi.tidal agent, as approved by the acquiring activity (see 6.2).

3.3.3 Protective treatment. When materials are used in the construction of the reducers that are subject to deterioration when exposed to enviromental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be

3.4 Standard parts. MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the manufacturer's drawings by their part numbers. Parts which are not covered by specifications or military standard, or which are not specifically described herein or on the drawings, shall be completely described on manufacturer's drawings.

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3.5 Design and construction.

3.5.1 Type I. The reducer shall be of the type suitable in all respects for use in aircraft oxygen systems and shall operate at supply pressures of 250 to 1800 psig. The outlet pressure shall be a predetermined fixed pressure within the range of 80 ± 10 psig for flow rates from zero to 140 1pm {NTP).

3.5.1.1 Mounting. The reducer shall be designed for permanent mounting on the airplane console. The location of the mounting holes shall have sufficient clearance between the mounting and inlet and outlet ports to allow for easy connection of fittings.

3.5.1.2 Shut-off valve. The reducer shall be provided with a shut-off valve which shall be capable of allowing the flow to be turned on or off. The shut-off valve shall be provided with a handle so that the valve may be actuated with one hand.

3.5.1.3 Ports. The reducer shall be provided with l/8-inch internal pipe threads in the inlet, relief, and outlet ports. Ports shall be clearly visible and functions permanently marked.

3.5.2 Type II, The reducer shall be capable of reducing emergency oxygen between the pressures of 250 and 1800 psig to a predetermined fixed pressure within the range Of 45 to 80 psig for flow rates from zero to 140 1pm (NTP).

3.5.2.1 Emergency supply release assembly shut-off valve. The emergency reducer shall be provided with a manual and automatic emergency supply release assembly. The valve shall be designed so that it may be actuated with one hand. The assembly shall consist of a common pin or disc attached to two pull cable assemblies. One pull cable assembly, for manual release, shall be attached to a ball handle; the other pull cable, for automatic release, shall be attached to the aircraft structure. The pin or disc shall be provided with a device to prevent accidental actuation of the bailout supply. The emergency supply release assembly when engaged shall prevent operation of the reducer.

3.5.2.2 Ports. The reducer shall be provided with l/8-inch internal pipe-threads in the relief and outlet ports. The inlet external threads shall be as specified by the acquiring activity (see 6.2). Ports shall be clearly visible and functions, permanently marked.

3.5.3 Types I and II. The reducer shall not induce or sustain hum or vibration within its operating range of pressure or flow, nor shall it sustain hum induced by aircraft vibration.

3.5.3.1 Filter. A triple filter consisting of 200 mesh wire screen or equivalent shall be firmly secured in the high pressure oxygen inlet of the reducer. This unit shall be easily removable for cleaning and replacement.

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3.5.3.2 Orientation. The reducer shall operate satisfactorily in any position in which the reducer is mounted (see 4.5.3).

3.5.3.3 Antiseize tape. Antiseize tape shall be used-on all male pipe thread fittings. Antiseize tape shall conform to and shall be applied as specified in MIL-T-27730. Tape shall not be used on flare tube fitting straight threads, coupling sleeves, or on the outer side of tube flares. None of the tape shall be allowed to enter the inside of a fitting.

3.5.3.4 Decreasing. Prior to assembling, all internal surfaces of the reducer shall be degreased by flushing with a cleaning compound conforming to MIL-C-81302 or by using a vapor phase degreaser in accordance with MIL-T-81533. Components shall be cleaned by immersing, scrubbing or pressure spraying with MIL-C-81302 cleaning compound or ultrasonics may be used in conjunction with vapor decreasing or MIL-C-81302 cleaning compound. After completion of the cleaning and when assembled, a General Electric Type H leak detector or equivalent halide testing apparatus shall be used to determine the absence of the cleaning compound.

3.6 Performance.

3.6.1 Leakage. The reducer, when tested as specified in 4.6.2, shall not show any evidence of leakage.

3.6.2 Pressure and flow.

3.6.2.1 Type I, The outlet pressure of the reducer, when tested as specified in 4.6.3,1, shall at no time be less than 70 psig nor greater than 90 psig.

3.6.2.2 Type II. The outlet pressure of the reducer, when tested as specified in 4.6.3.2, shall at no time be less than 45 psig nor greater than 80 psig,

3.6.3 Shut-off valve.

3.6.3.1 Type I. The shut-off valve on the reducer, when tested as specified in 4.6.4.1, shall permit a minimum flow of 140 1pm (NTP) in the “on" position and shall not allow any flow in the "off" position. The maximum force required to turn the valve on or off shall be 15 inch pounds.

3.6.3.2 Type II. The emergency supply release shut-off valve on the reducer, when tested as specified in 4.6.4.2, shall permit a minimum flow of 140 1pm (NTP) when actuated and shall not allow any flow when engaged. The pull force required to actuate the emergency supply release assembly shall be a minimum of 10 pounds and a maximum of 28 pounds.

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3.6.4 Pressure relief. The pressure relief valve, when tested as specified in 4.6.5, shall open at 120 ± 10 DSig and shall allow a mininimum flow of 80 lpm at 130 psig through the valve. The pressure relief valve shall reseat at 110 psig and show no evidence of leakage.

3.6.5 Proof pressure. The reducer, when tested as specified in 4.6.6, shall show no evidence of leakage or distortion.

3.6.6 High and low temperature operation. The reducer, when tested as specified in 4.6.7 shall operate without failure and shall not show any evidence of mechanical or material failure,

3.6.7 Vibration. The reducer, when tested as specified in 4.6.8, shall not show any evidence of mechanical or material failure.

3.6.8 Endurance (Type I). The on-off shut off valve, when tested as specified in 4.6.9, shall operate satisfactorily.

3.6.9 Oxygen bomb test. The reducer, when tested as specified in 4.6.10, shall not ignite and there shall not be any evidence of charring or deterioration.

3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.8 Dimensions.

3.8.1 Type I. The reducer shall have maximum overall dimensions of 4-1/4 by 1-1/2 by 1-1/2 inches.

3.8.2 Type II. The reducer shall have maximum overall dimensions of 3 by 2-1/4 by 1-1/2 inches.

3.9 Weight. The weight of the pressure regulating valve shall riot exceed 0.4 pounds.

3.10 Threads.

3.10.1 Screw. Machine screw threads shall be in accordance with MIL-S-7742.

3.10.2 Pipe. Pipe threads shall be in accordance with MIL-P-7105.

3.11 Finish. Aluminum and aluminum alloys shall be anodized in accordance MIL-A-8625 or MIL-C-5541.

3.12 Identification of product. The valves shall be marked for identification in accordance with MIL-STD-130, except that the National Stock Number shall be omitted from the nameplate.

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3.12.1 Nameplate. The following information shall be provided on the nameplate:

REDUCER, OXYGEN PRESSURE
MIL-R-17852C
Contract or Order No.
Manufacturer's Part No.
Manufacturer's Name or Trade Mark

3.13 Workmanship. The reducer shall be uniform in quality and shall be free from irregularities, defects or foreign matter which could adversely affect safety, performance, reliability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services- conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item meets the requirements of this specification (see 4.3).

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.4.

4.3 First article inspection. First article inspection shall consist of the following examinations and tests:

Inspection	Test method
Visual examination	4.6.1
Dimensions	4.6.1.1.1
Leakage	4.6.2
Pressure and flow	4.6.3

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Shut-off valve	4.6.4
Pressure relief	4.6.5
Proof pressure	4.6.6
High and low temperature operation	4.6.7
Vibration	4.6.8
Endurance (type I)	4.6.9
Oxygen bomb test	4.6.10

4.3.1 First article samples. Unless otherwise specified, as soon as practicable, after the award of the contract or order, the manufacturer shall submit two reducrers. he samples shall be representative of the construction workmanship, components and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2). Approval of the first article samples or the waiving of first article inspection does not exclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government-as directed by the contracting officer (see 6.2).

4.3.1.1 First article information. Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. One approved reducer will be returned to the manufacturer for use in monitoring production. The other sample will be consumed or destroyed in the first article inspection and will not be considered as part of the quantity to be delivered under contract.

4.4 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. Quality conformance inspection shall be as specified in table I.

4.4.1 Sampling.

4.4.1.1 Inspection lot.

4.4.1.1.1 Reducer. An inspection lot size shall be expressed in units of reducers, made essentially under the same conditions and from the same materials and components, The sample unit shall be one reducer.

4.4.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing reducers of one type, fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing reducers of one type, fully prepared for delivery, with the exception that it need not be sealed.

4.4.1.2 Sampling for tests and examinations of reducers. The sample size, acceptance criteria, tests and examinations required for the reducers shall be as specified in table I.

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4.5 Test condition.

4.5.1 Temperature and pressure. Unless otherwise specified, tests shall be conducted at local ambient temperature arid barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, where required, to normal temperature and pressure (NTP) conditions. NTP conditions are 29.92 inches of mercury (101.3kPa) and 21°C (70°F). Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.5.2 Gas. Unless otherwise specified, the gas used in testing the reducers sha11 be oxygen conforming to MIL-O-27210, type I. When specified, nitrogen conforming to BB-N-411, type I, class I, grade B, or air conforming to BB-A-1034, may be used.

4.5.3 Orientation. When subjected to the test specified in 4.6.3, the reducers shall be positioned with the centerline of the passage parallel to a vertical plane. Each consecutive reducer tested shall be rotated 90 degrees and 180 degrees about an axis perpendicular to the vertical plane. The rotation shall always be in the same direction.

4.6 Inspection methods.

4.6.1 Visual examination.

4.6.1.1 Reducer. Every reducer shall be examined visually for critical defects to determine conformance to this specification. Every reducer, selected as a sample unit from the lot, shall be visually examined for minor defects to determine conformance to this specification. The classification of defects, table II, shall be used to classify the defects found.

4.6.1.1.1 Dimensions. The reducers shall be checked dimensionally to determine conformance to the dimensions specified herein.

4.6.1.2 Packaging. Each of the fully prepared shipping containers, containing reducers of one type, selected as a sample unit from the lot, shall be examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects, table III, shall be used to enumerate the defects found.

4.6.2 Leakage. The outlet of the reducer shall be sealed with a suitable plug. The reducer shall be actuated and a pressure of 1800 psig shall then be applied to the inlet of the reducer and maintained for a period of 15 minutes. The reducer shall pass the requirements specified in 3.6.1.

4.6.3 Pressure and flow.

4.6.3.1 Type I. The reducer shall be preset at 80 psig and the inlet connected to an oxygen pressure source. The reducer shall then be subjected to an inlet pressure of 2400 psig and the flow at the reducer

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outlet shall be varied from zero to 140 1pm (NTP), in increments of 20 1pm. The outlet pressure shall be recorded at the various flow rates. The test procedure shall be repeated utilizing inlet pressures of 1800, 1000, 500, and 250 psig. The reducer shall pass the requirements specified in 3.6.2.1.

4.6.3.2 Type II. The reducer shall be preset at 80 psig and the inlet connected to an oxygen pressure source. The test procedure specified in 4.6.3.1 shall be repeated utilizing inlet pressures cf 1800, 1000, 500 and 250 psig. The reducer shall pass the requirements specified in 3.6.2.2.

4.6.4 Shut-off valve.

4.6.4.1 Type I. The reducer shall be connected to an oxygen pressure source and the reducer subjected to an inlet pressure of 250 psig. The shut-off valve shall then be turned to the "on" position, and the flow recorded. The inlet pressure shall then be increased to 2400 psig. The reducer shall be turned to the "off" position and the flow recorded. The force required to open and close the valve shall also be noted. The reducer shall pass the requirements specified in 3.6.3.1.

4.6.4.2 Type II. The reducer shall be connected to an oxygen pressure source and the reducer subjected to an inlet pressure of 250 psig. The emergency supply release assembly shall be actuated and the flow recorded. The inlet pressure Shall then be increased to 1800 psig. The pull force required to actuate the emergency supply release assembly shall first be measured with zero pressure applied to the outlet of the reducer and again, after resetting the reducer, with 90 psig from an external source of oxygen applied to the outlet of the reducer. The release assembly shall be engaged and the flow measured with an inlet pressure of 1800 psig. The reducer shall pass the requirements specified in 3.6.3.2.

4.6.5 Pressure relief. A pressure of 1800 psig shall be applied to the inlet of the pressure regulating valve. An increasing pressure shall then be applied to the outlet of the valve until the relief valve opens. The relief valve opening pressure shall be noted. The relief valve pressure shall then be adjusted to 130 psig and the flow rate through the relief valve shall be noted. The valve outlet pressure shall then be decreased until the relief valve closes. The relief valve closing pressure shall be noted. The valve shall pass the requirements specified in 3.6.4.

4.6.6 Proof pressure. The reducer shall be connected to an oxygen pressure source and the reducer subjected to an inlet pressure of 3000 psig. The 3000 psig pressure shall be maintained for a period of 3 minutes and the reducer shall pass the requirements specified in 3.6.5.

4.6.7 High and low temperature operation. A pressure of 1800 psig shall be applied to the inlet of the reducer. The reducer shall then be subjected to a temperature of 160 ± 5°F (71 ± 2.8°C) for 3 hours.

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After the conditioning period and while still at this temperature, the reducer shall be subjected to and pass the pressure and flow test. The reducer shall then pass the requirements specified in 3.6.5. The above test procedure shall be repeated utilizing a temperature of -65 ± 5°F (-54 ± 2.8°C) for 3 hours. After the conditioning period and while still at this temperature the reducer shall be subjected to and pass the pressure and flow test. The reducer shall then pass the requirements specified in 3.6.6.

4.6.8 Vibration. A pressure of 1800 psig shall be applied to the inlet of the reducer. The reducer shall then be vibrated for 3 hours at a frequency from 300 to 3000 cycles per minute and a double amplitude of 0.018 to 0.020 inches. The reducer shall be cycled six times, each cycle being 30 minutes and consisting of varying frequency from 300 to 3000 and back to 300 cycles per minute. The reducer shall be mounted in each of the three mutually perpendicular axes of vibration for a one hour period in each plane. During the vibration period, a flow of 30 ± 5 liters per minute shall be maintained through the reducer. The reducer shall pass the requirements specified in 3.6.7. The reducer shall then be subjected to and pass the leakage, pressure, and flow tests.

4.6.9 Endurance (Type I). The actuating handle shall be manually operated 1000 times. The reducer shall pass the requirements specified in 3.6.8.

4.6.10 Oxygen bomb test. The reducer shall be subjected to the oxygen bomb test to determine compatibility of the item with high pressure oxygen.

4.6.10.1 Equipment, The equipment used in performing these tests shall be in accordance with Figure 1. The main valve shall be a rapid-opening 1/4 inch valve, shall require no internal lubrication for operation, and shall be rated for at least 3,000 psig operation. The valve shall be leak tight when closed and shall open with such rapidity that at least 95 percent of the total pressure rise shall have occurred within the test chamber in 20 ± 7 milliseconds. The pressure rise time may be determined by measurement with a fast-responding small volume pressure transducer substituted for the reducer in the test setup. The tubing and fitting connecting the valve to the test chamber shall be 3/16 inch size and shall be fabricated from corrosion-resistant steel. The run of tubing between the valve and the test chamber shall not be. bent more than 4 degrees. The bleed valve shall be of a type which uses no organic material to accomplish its seal and requires no interior lubrication for operation. The bleed valve shall be installed so that when it is closed, the test chamber does not communicate with the packing side of the valve.

4.6.10.2 Procedure (reducer). The outlet connections of the reducer shall be sealed. The reducer inlet shall be installed in the test setup shown in Figure 1. The quick-acting valve shall be fully opened and the reducer pressurized with oxygen to a pressure of 2000 psig. This pressure shall be maintained for 30 seconds and then released through the bleed

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valve. The reducer shall be disconnected and examined for indication of charring or deterioration of any nonmetallic material. The valve shall pass the requirements specified in 3.6.9.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Unless otherwise specified, each reducer shall be cleaned for oxygen service in accordance with industry practice and 3.5.3.4 and then packaged in accordance with MIL-P-116, Method IC-1. Contact preservatives is prohibited. Each reducer shall be sealed in a bag of MIL-B-22191, Type II material, cushioned to prevent damage and packaged in a waterproof fiber can conforming to MIL-C-3955, Type I or II, Grade B.

5.1.1.2 Intermediate packaging. Six unit packs shall be packaged within a fiberboard container conforming to PPP-B-636, Type I, Weather Resistant Class, Grade V3c or V3s. The manufacturer's seam shall be adhesive sealed. All seams and joints shall be sealed with tape conforming to PPP-T-60.

5.1.2 Level C. The reducer shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified, sixty reducers, packaged as specified in 5.1.1, shall be packed in overseas type exterior containers conforming to PPP-B-591, PPP-B-636 or PPP-B-640, weather resistant class. Closure and strapping of the containers shall be in accordance with the applicable container specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight limitations of the applicable container specification.

5.2.2 Level B. Sixty reducers, packaged as specified in 5.1.1, shall be packed in domestic type exterior containers conforming to PPP-B-591, PPP-B-636, or PPP-B-640, Domestic Class. Closure and strapping of the containers shall be in accordance with the applicable container specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight limitations of the applicable container specification.

5.2.3 Level C. The packaged reducers, which the carrier requires to be packed, shall be packed within exterior type shipping containers in a manner that shall insure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

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5.3 Marking. Nomenclature tags shall be inserted in each bag showing title, part number or other data as required by acquiring activity. In addition, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking. The following precautionary marking shall appear on each package:

"CAUTION: DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT THE OXYGEN PRESSURE REDUCERS"

6. NOTES

6.1 Intended use,

6.1.1 Type I. The oxygen pressure reducer is intended for use in aircraft where it is desired to reduce oxygen at pressures between 250 and 1800 psig to a pressure range of 80 ± 10 psig at flow rates up to 140 1pm (NTP).

6.1.2 Type II. The emergency oxygen pressure reducer is intended for use in bailout systems where it is desired to reduce oxygen at pressures between 250 to 1800 psig to a pressure range of 45 to 80 psig at flow rates up to 140 1pm (NTP).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type required (see 1.2).

c. Fungicidal agent to be used (see 3.3.2.2).

d. Size of thread required (see 3.5.2.2).

e. Whether first article inspection is required and where the sample should be delivered (see 4.3.1).

f. Selection of applicable levels of preservation, packaging and packing.

g. Applicable methods of cleaning and preservation.

h. Items of data required (see 6.3).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into

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the contract. When the provisions of DAR-7-104.9(n) are not invoked; the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.3.1.1	First Article Inspection Reports	DI-T-5329 - Inspection Test Reports

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

6.4 First article. When a first article is required, it shall be tested under the appropriate provision of 7-104.55 of the Defense Acquisition Regulation. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangements for examinations, tests and approval of the first article.

Preparing activity:
Navy - AS
(Project No. 1660-N433)

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TABLE I. Sample size, acceptance criteria, tests and examinations of the reducers.

Inspection	Method	Sample size	Acceptance criteria
Visual examination (see classification of defects)	4.7.1.1	Every reducer for critical defects. Inspection level II for minor defects 1/	Reject all units with any critical defect. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensions	4.6.1.1.1	Inspection level S-2 1/	Acceptance number zero, rejection number 1.
Leakage	4.6.2	Every reducer	Reject all defective units.
Pressure and flow	4.6.3	Every reducer	Reject all defective units.
Shut-off valve	4.6.4	Every reducer	Reject all defective units.
Pressure relief	4.6.5	Every reducer	Reject all defective units.
Packaging	4.6.1.2	Inspection level S-2	Total acceptable quality level of 4.0 percent defective.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

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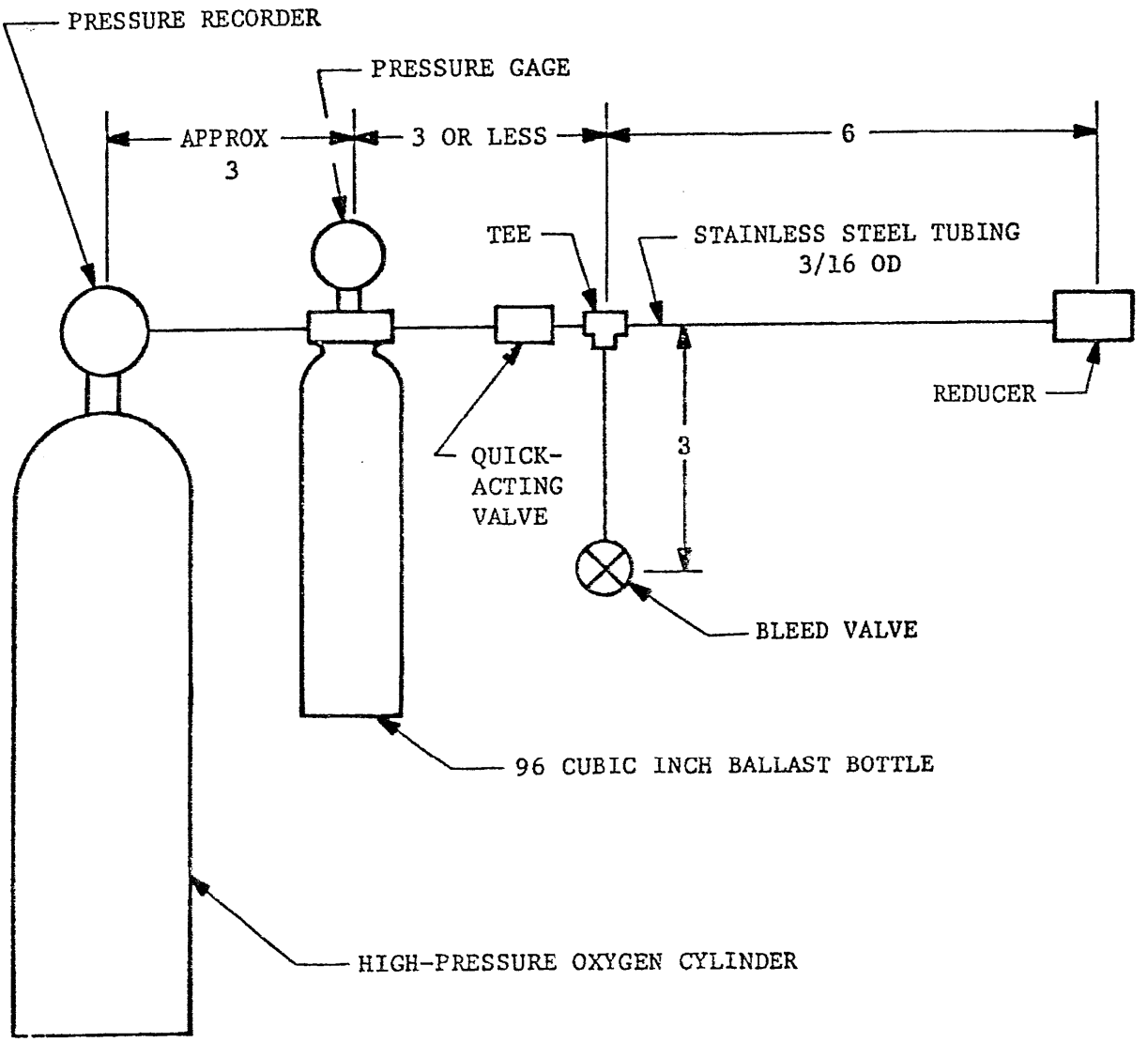
TABLE II. Classification of defects for visual examination of the reducers.

Critical	Minor
1. Material imperfections - foreign matter embedded.	201. Marking - missing, insufficient, incorrect, illegible or not permanent.
2. Surface - unclean, rough, misaligned, or containing cracks, nicks or other flaws.	
3. Any component missing, malformed, fractured or otherwise damaged.	
4. Any component loose or otherwise not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	

TABLE III. List of defects for packaging

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements; more than one type in the same container.

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DIMENSIONS IN INCHES.

FIGURE 1. Oxygen bomb equipment.

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